On Ordowski's characterization of odd primes by the number of squares modulo n (OEIS A000224): even numbers, prime powers, numbers $p^a q^b$, and products of three distinct primes, formalized in Lean 4

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October 12, 2026

Abstract

Let $a(n)$ be the number of squares modulo n (OEIS A000224 [1]). T. Ordowski conjectured that for $n > 1$, $n^2 \equiv 1 \pmod{a(n)(a(n)-1)}$ holds if and only if n is an odd prime. **The full conjecture remains open.** We prove it for every $n > 1$ that is even, a prime power p^k ($k \geq 1$), a number $p^a q^b$ with $p \neq q$ odd primes and $a, b \geq 1$ (in particular every pq), or a product pqr of three distinct odd primes. The prime-power case uses Stangl's formula for $a(p^k)$ [3], which we also formalize. The semiprime case combines a size bound with a divisibility bound and a finite check. The three-prime case bounds the smallest prime by a size argument, views the congruence as a quadratic in the largest prime to bound the middle one, and finishes with a kernel-checked finite verification of about 9.4×10^4 cases. The case $p^a q^b$ uses Stangl's formula to write the congruence as a quadratic in p^a and in q^b; size bounds give $Q \leq 40$ and $p \leq 67$, explicit bounds on q and on $\min(p^a, q^b)$ follow, and a kernel-checked integer-root search finishes the proof. Everything is formalized in Lean 4 with Mathlib against the exact definition of a in Google DeepMind's `formal-conjectures` repository [2]. The development uses no `sorry` and no `native_decide`, and only the standard axioms.

1 Introduction

For $n \geq 1$ let $a(n) = \#\{k^2 \bmod n : 0 \leq k < n\}$.

Conjecture 1 (Ordowski, 2025)**.** *For $n > 1$, n is an odd prime if and only if $n^2 \equiv 1 \pmod{a(n)(a(n)-1)}$.*

The conjecture appears as a comment on A000224 [1], together with M. Marcus's remark that it holds up to $n = 10^8$. It is formalized as `OeisA224.conjecture` in `FormalConjectures/OEIS/224.lean` of `formal-conjectures` [2], where it is tagged `research open`. The forward direction is immediate from $a(p) = (p+1)/2$. The open part is to show that no composite n satisfies the congruence.

2 Related work

The count $a(n)$ is multiplicative, and Stangl [3] gives its value at prime powers; see also Finch and Sebah [4]. On October 12, 2026 we searched OEIS, the web, MathOverflow and the issues and pull requests of `formal-conjectures` (upstream commit `34ee0e3`). We found no proof of Conjecture 1,

partial or complete. The cases treated here therefore appear to be new, although the arguments are elementary. The formalization uses Lean 4 [6] and Mathlib [5].

3 Statement

Theorem 2. *Conjecture 1 holds for every $n > 1$ that is (i) even, (ii) a prime power p^k with $k \geq 1$, or (iii) of the form $p^a q^b$ with $p \neq q$ odd primes and $a, b \geq 1$ (this includes pq), or (iv) a product pqr of distinct odd primes p, q, r.*

The cases not covered are exactly the odd n with at least three distinct prime factors, other than squarefree pqr (for example $pqrs$ or p^2qr).

4 Proof overview

4.1 Preliminaries

Lemma 3. *$a(n) = \#\{x^2 : x \in \mathbb{Z}/n\}$; a is multiplicative; and $a(p) = (p+1)/2$ for odd primes p.*

Proof. The first claim holds because reduction mod n is a bijection $\{0, \ldots, n-1\} \to \mathbb{Z}/n$. If $\gcd(m, n) = 1$, the CRT ring isomorphism $\mathbb{Z}/mn \cong \mathbb{Z}/m \times \mathbb{Z}/n$ commutes with squaring, so it maps the set of squares onto the product of the two sets of squares. For an odd prime p, the squares are the values k^2 with $0 \leq k \leq (p-1)/2$, since $x^2 = (-x)^2$. These values are distinct: $k^2 = l^2$ forces $p \mid (k-l)(k+l)$, and $0 \leq k + l < p$. □

4.2 The even case

If n is even, then $n^2 - 1$ is odd, while $a(n)(a(n) - 1)$ is a product of two consecutive integers and hence even. So $n^2 \not\equiv 1$. (If $a(n)(a(n)-1) = 0$, the congruence would say $n^2 = 1$, which is impossible for $n > 1$.)

4.3 The semiprime case

Theorem 4. *Let $p < q$ be odd integers with $p \geq 3$ prime, and let $s = (p+1)(q+1)/4$. Then $s(s-1) \nmid (pq)^2 - 1$.*

Primality of q is not needed. By Lemma 3, $a(pq) = s$ for distinct odd primes, so Theorem 4 gives the semiprime case of Theorem 2.

Proof. Put $P = (p+1)(q+1)$, so that $16\, s(s-1) = P(P-4)$. Suppose $Q\, s(s-1) = (pq)^2 - 1$, so $Q\, P(P-4) = 16((pq)^2 - 1)$.

Step 1: $Q \leq 15$. Expanding, $P(P-4) - ((pq)^2 - 1) = 2pq(p+q+1) - 4pq + (p+q+1)^2 - 4(p+q+1) + 1 > 0$. Hence $Q < 16$.

Step 2: $p \leq 61$. By Step 1, $16((pq)^2 - 1) \leq 15P(P-4) < 15P^2$. Suppose $p \geq 62$. Since $q > p$ we have $(q+1)p \leq (p+1)q$, so $P^2p^2 \leq (p+1)^4q^2$. Also $15(p+1)^4 \leq 16p^4 - 16$ for $p \geq 62$. Together with $q \geq p$ this gives $15P^2 \leq 16((pq)^2 - 1)$, a contradiction.

Step 3: $q + 1 \mid 16(p^2 - 1)$. Since $q + 1 \mid P$, we get $q + 1 \mid 16((pq)^2 - 1)$. Also $16((pq)^2 - 1) - 16(p^2 - 1) = 16p^2(q^2 - 1)$ is divisible by $q + 1$.

Step 4: finite check. For each prime $p \in [3, 61]$ and each divisor $d = q+1$ of $16(p^2-1)$ with $q > p$ odd, direct computation shows $s(s-1) \nmid (pq)^2 - 1$. There are only finitely many such pairs. □

4.4 Three distinct odd primes

Theorem 5. *Let $3 \le p < q < r$ be integers with p prime, and let s satisfy $8s = (p+1)(q+1)(r+1)$. Then $s(s-1) \nmid (pqr)^2 - 1$.*

Primality of q and r is not needed. By Lemma 3, $a(pqr) = s$ for distinct odd primes, which gives case (iv) of Theorem 2.

Proof. Let $n = pqr$ and $P = (p+1)(q+1)(r+1) = 8s$, and suppose $Q\,s(s-1) = n^2 - 1$. Then $Q\,P(P-8) = 64(n^2-1)$.

Step 1: $1 \le Q \le 63$. Since $P - 8 \ge n$, we have $P(P-8) > n^2 - 1 > 0$.

Step 2: $p \le 400$. From $p \le q \le r$ we get $p^3 P \le (p+1)^3 n$. Combined with $64(n^2-1) \le 63P^2$ this gives $(64p^6 - 63(p+1)^6)n^2 \le 64p^6$. For $p \ge 401$ the bracket is at least 65, and $n^2 > p^6$, a contradiction.

Step 3: a quadratic in r. Put $A = (p+1)(q+1)$ and $B = pq$. Expanding the identity gives

$$(64B^2 - QA^2)\,r^2 = (2QA^2 - 8QA)\,r + (QA^2 - 8QA + 64),$$

where both coefficients on the right are positive. Hence $c_2 := 64B^2 - QA^2 > 0$. With $K = Q(p+1)^2$ and $d = 64p^2 - K$ we have $c_2 = dq^2 - K(2q+1)$, so $d > 0$. Moreover $c_2 r \le c_1 + c_0 < 3QA^2$ and $q < r$, so $c_2 q < 3K(q+1)^2$. If $q \ge 8$ and $dq \ge 6K$, then $c_2 q \ge 4Kq^2 - Kq > 3K(q+1)^2$, a contradiction. Therefore $q \le 6K/d + 8$. The near-misses $Q(1+1/p)^2 = 64$, namely $(p, Q) = (3, 36)$ and $(7, 49)$, have $d = 0$ and are excluded.

Step 4: finite check. For each prime $p \le 400$, each $Q \le 63$ with $d > 0$, and each $q \le 6K/d + 8$ with $q > p$ (about 9.4×10^4 triples), the quadratic in Step 3 has no natural root $r > q$. A root would equal $(c_1 + c)/(2c_2)$, where $c = \sqrt{c_1^2 + 4c_2 c_0}$. The check proposes c by Newton's method and verifies $c^2 \le D < (c+1)^2$ directly, so the correctness of Newton's method is never used. □

4.5 Odd prime powers

Lemma 6 (Stangl [3])**.** *For an odd prime p and $k \ge 2$, $a(p^k) = a(p^{k-2}) + \frac{1}{2}p^{k-1}(p-1)$. Consequently $2(p+1)\,a(p^k) = p^{k+1} + p + 2$ if k is even and $p^{k+1} + 2p + 1$ if k is odd.*

Proof. Let $N = p^k$. A square $x^2 \bmod N$ is divisible by p iff $p \mid x$. Writing $x = pt$ gives $(pt)^2 \bmod p^2 M = p^2(t^2 \bmod M)$ with $M = p^{k-2}$, so the squares divisible by p are exactly p^2 times the squares modulo M. For the squares prime to p: if $x, y \in [0, N/2)$ are prime to p and $N \mid (y-x)(y+x)$, then p cannot divide both factors (as p is odd and $p \nmid y$), so N divides one factor, forcing $x = y$. Replacing x by $N - x$, every unit square has such a representative. Hence the unit squares number $\varphi(N)/2$. The closed form follows by induction from $a(1) = 1$ and $a(p) = (p+1)/2$. □

Theorem 7. *If p is an odd prime, $k \ge 2$, and s satisfies the identity of Lemma 6, then $s(s-1) \nmid p^{2k} - 1$. Consequently, $(p^k)^2 \not\equiv 1 \pmod{a(p^k)(a(p^k)-1)}$, and Conjecture 1 holds for all odd prime powers.*

Sketch. Let $X = p^k$ and $M = 2(p+1)$, and write $Q\,s(s-1) = X^2 - 1$. Multiplying by M^2 gives $Q(Ms)(Ms - M) = M^2(X^2 - 1)$. For even k, reducing modulo p gives $p \mid 4(p+1)^2$, which is impossible. For odd $k \ge 3$, a size estimate gives $1 \le Q \le 7$, and reducing modulo p gives $p \mid Q - 4$. If $Q = 4$, the equation becomes $(2p+1)X^2 = 2p^2 X + p^2$, which is impossible since $X > p$. Otherwise $p = 3$ and $Q \in \{1, 7\}$, and both cases are ruled out by elementary inequalities. □

4.6 Two distinct primes with arbitrary exponents

Theorem 8. *Let $p < q$ be distinct odd primes and $a, b \geq 1$. Then $n = p^a q^b$ does not satisfy $n^2 \equiv 1 \pmod{a(n)(a(n)-1)}$.*

Sketch. Write $X = p^a$, $Y = q^b$, $s = a(n) = a(X)a(Y)$ and $M = 4(p+1)(q+1)$. By Lemma 6, $U := 2(p+1)a(X) = pX + c$ and $V := 2(q+1)a(Y) = qY + e$ with $c \in \{p+2, 2p+1\}$ and $e \in \{q+2, 2q+1\}$, so $Ms = UV$. Suppose $Q\,s(s-1) = n^2 - 1$. Then Q is prime to n (it divides $n^2 - 1$), and

$$Q\,UV(UV - M) = M^2(X^2Y^2 - 1). \tag{$*$}$$

Step 1: $1 \leq Q \leq 40$. Expanding, $UV - M \geq pq\,n$ (the term $qYc \geq q^2(p+2)$ already exceeds M), so $Q\,p^2q^2n^2 \leq M^2(n^2-1)$ and $Qp^2q^2 < M^2 = 16(p+1)^2(q+1)^2$. Since $8pq \geq 5(p+1)(q+1)$ for $p \geq 3$, $q \geq 5$, this forces $Q \leq 40$.

Step 2: $p \leq 67$. If $p \geq 71$ then $pq/((p+1)(q+1)) \geq 5112/5256$, and comparing $(*)$ with $pqn \leq Ms \leq ((p+1)(q+1))^2 n/(pq)$ gives $15 < Q < 17$. But $Q = 16$ means $(4s-2)^2 = n^2 + 3$, impossible for $n \geq 2$. The primes $68 \leq p \leq 70$ do not exist.

Step 3: two quadratics. Substituting $U = pX + c$ in $(*)$ gives

$$(M^2Y^2 - Qp^2V^2)X^2 = QpV(2cV - M)\,X + \big(QcV(cV - M) + M^2\big),$$

and symmetrically a quadratic in Y with leading coefficient $M^2X^2 - Qq^2U^2$. Because $cV > M$ and $eU > M$, both right-hand sides have positive coefficients, so both leading coefficients are positive.

Step 4: bounding q. Let $E = 16(p+1)^2 - Qp^2$ (never 0, since $p \nmid 16(p+1)^2$) and $G = 16(p+1)^2X^2 - QU^2$. The quadratic in Y together with $Y \geq q$ gives $q(-G) < 48(p+1)^2X^2$ if $G < 0$ and $qG \leq 13QU^2$ if $G > 0$. If $E < 0$ this yields $q \leq 48(p+1)^2/(-E)$. If $E > 0$ and $2Q(2p+1)(2pX+2p+1) \leq EX^2$ (automatic for $X > 2 \cdot 10^6$), then $2G \geq EX^2$ and $pU \leq (p+1)^2X$ give $q \leq 26Q(p+1)^4/(p^2E)$. The finitely many remaining $X \leq 2 \cdot 10^6$ are treated one at a time with the X-dependent bound.

Step 5: bounding $\min(X, Y)$. Write $\Delta = M^2 - Qp^2q^2 > 0$. Expanding $(*)$ with $U^2 - p^2X^2 \leq 9p^2X$ and $V^2 - q^2Y^2 \leq 9q^2Y$ gives $\min(X,Y)\,\Delta < 45Qp^2q^2 + M^2$.

Step 6: finite check. For every prime $p \leq 67$, every $1 \leq Q \leq 40$ with $p \nmid Q$, every odd $q > p$ under the bound of Step 4, and every power $X = p^a$ (resp. $Y = q^b$) under the bound of Step 5 (or $X \leq 2 \cdot 10^6$ in the small-X branch), and for both admissible values of e (resp. c), the quadratic in the other variable has no natural root divisible by q (resp. p). Roots are tested as in Theorem 5: a Newton candidate c is verified by $c^2 \leq D < (c+1)^2$. □

The case $a = b = 1$ is covered as well, so Theorem 8 contains the semiprime case. An independent search up to $n \leq 10^{15}$ also found no counterexample of this shape.

5 Formalization

All declarations are in namespace `OeisA224`. The definition `a` in `Defs.lean` is copied verbatim from `formal-conjectures`. The theorems `conjecture_main`, `conjecture_main_three` and `conjecture_main_ppq` conclude exactly `(n.Prime ∧ n ≠ 2) ↔ n ^ 2 ≡ 1 [MOD a n * (a n - 1)]` under its case hypothesis.

Lean name (file)	Statement
`a_eq_aZ`, `aZ_mul`, `aZ_prime` (`Count.lean`)	Lemma 3
`even_case`, `prime_direction` (`Basic.lean`)	even case; odd prime ⇒ congruence
`finCheck_true` (`FiniteCheck.lean`)	Step 4, checked with `decide +kernel`
`pq_reduction`, `pq_case`, `semiprime_case` (`Semiprime.lean`)	Theorem 4 and case (iii)
`pp_even`, `pp_odd` (`PrimePowerCore.lean`)	arithmetic core of Theorem 7
`stangl`, `prime_power_case` (`PrimePower.lean`)	Lemma 6; Theorem 7
`conjecture_of_even_or_prime_or_semiprime`, `conjecture_main` (`Main.lean`)	Theorem 2, cases (i)–(iii)
`finCheck3_true` (`FiniteCheck3.lean`)	Step 4 of Theorem 5, `decide +kernel`, in 10 chunks
`pqr_Q`, `pqr_p_le`, `pqr_quad`, `pqr_bounds`, `rootFree_spec`, `pqr_no`, `three_primes_case` (`ThreePrimes.lean`)	Theorem 5
`three_primes_case'`, `conjecture_main_three` (`MainThree.lean`)	Theorem 2, cases (i), (ii), (iv) and pq
`core_quadX`, `core_quadY`, `core_qbound`, `core_min`, `core_large` (`TwoPrimePowers.lean`)	Steps 2–5 of Theorem 8
`okP2_of_mem` (`FiniteCheckPQ*.lean`, `FCPQ/*.lean`)	Step 6 of Theorem 8, `decide +kernel`, split into 45 modules
`noY_one`, `noX_one` (`TwoPrimePowersGlue.lean`)	soundness of the root tests
`KM_lt`, `Q_le_40`, `triple_spec`, `smallCheck_spec`, `ppq_finite`, `two_prime_powers_case`, `conjecture_main_ppq` (`TwoPrimePowersMain.lean`)	Step 1, assembly, Theorem 8, Theorem 2 (i)–(iv)

Table 1: Theorems and their Lean names.

6 Reproducibility

- Source code: `https://github.com/het4rk/ordowski-a224`.
- Toolchain: `leanprover/lean4:v4.35.0-rc4`. Mathlib: tag `v4.35.0-rc4`, commit `1f414401f69059aa7eead47b53ee40bd38455eeb`, pinned in `lake-manifest.json`.
- Build:

```
lake exe cache get
lake build
lake env lean Axioms.lean
```

`FiniteCheck.lean` and `FiniteCheck3.lean` each take about 4–5 minutes and several GB of RAM; the 45 modules of the $p^a q^b$ check take roughly 20 minutes in total and should be built one at a time (as CI does).

- Axioms: all 18 declarations listed in `Axioms.lean`, including `conjecture_main_three` and `conjecture_main_ppq`, depend only on standard axioms: `propext`, `Classical.choice` and `Quot.sound` (`finCheck3_true` uses only `propext`).
- Attestation: `verification/` contains the fresh-clone build log, the axiom output and `VERIFICATION.md`, which records the commit, the toolchain and SHA-256 hashes of the Lean sources. CI (`https://github.com/het4rk/ordowski-a224/actions/workflows/lean.yml`) rebuilds the project on every push. It rejects `sorry` and non-whitelisted axioms, and uploads the build log and axiom report as artifacts.

7 Limitations

The full conjecture remains open. The untreated case is odd n with at least three distinct prime factors, other than squarefree pqr; for example $pqrs$ and p^2qr. For squarefree n with k factors, the argument of Theorem 5 gives $Q < 4^k$ and a bound on the smallest prime of order $k\,4^k$. For each fixed k an iterated version of Steps 2–4 reduces the problem to a finite search, but the search grows far too quickly to check in the kernel even for $k = 4$. A proof uniform in k would need a new idea. The independent computational checks in `scripts/` are not part of the formal proof.

References

[1] OEIS Foundation Inc., *A000224: Number of squares mod n*, `https://oeis.org/A000224`.

[2] Google DeepMind, *formal-conjectures*, `https://github.com/google-deepmind/formal-conjectures`, file `FormalConjectures/OEIS/224.lean`.

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[6] L. de Moura, S. Ullrich, *The Lean 4 theorem prover and programming language*, in CADE-28, LNCS 12699, Springer, 2021, 625–635.